

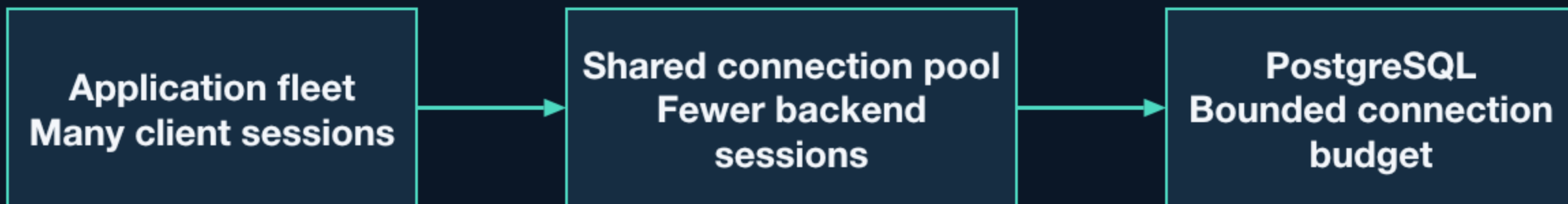
PGProxyScale

PostgreSQL connection sharing with explicit session-state control

Technical sales briefing

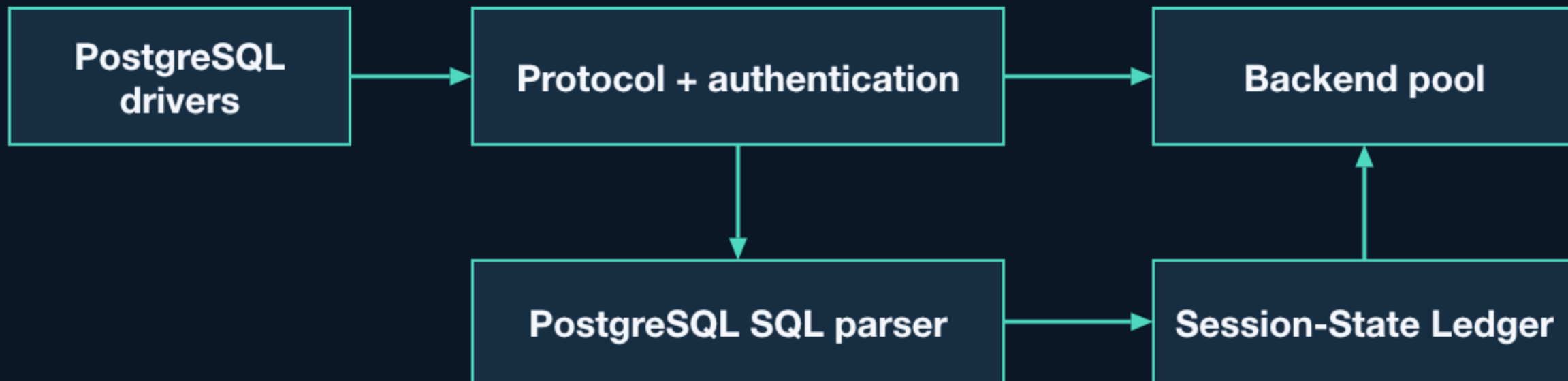
Design-partner preview / September 2026

The connection scale problem



Prepared statements and settings belong to sessions.
Sharing backends requires the proxy to preserve that state.

PGProxyScale architecture



Current: session and transaction modes, bounded messages and state replay

Planned: SQL policy, principal fairness, agent access and resilience

The ledger follows the client session



Release requires an idle backend, completed exchanges and no state that needs a dedicated connection.

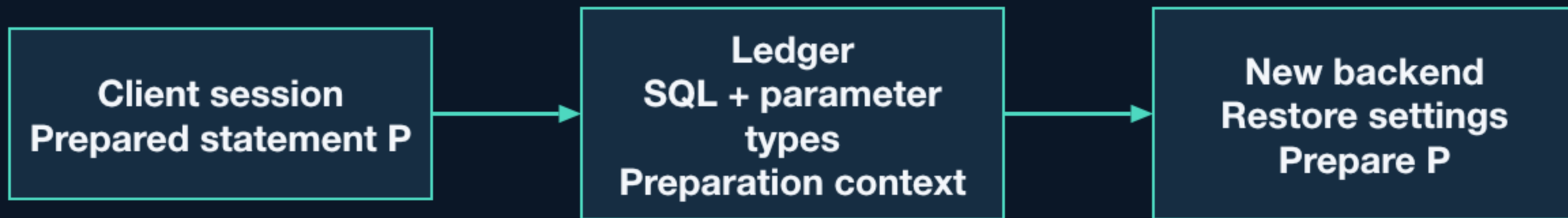
Before reuse: clear cancellation ownership and reset the backend.

Compatibility has an explicit state boundary

Session behavior	Current treatment	Connection sharing
SET / RESET and prepared statements	Track confirmed state and replay	Eligible when safe
Open transaction and active protocol batch	Keep backend ownership	Until exchange completes
LISTEN, held cursors, temp tables, session locks	Retain the backend	Reduced while state persists
Unknown session effects / changed parser context	Conservative retention	Requires workload review

Full state virtualization and zero pinning remain development goals.

Prepared statements across backend handoff



The client keeps its statement name while the proxy restores the required state before forwarding the next execution.

Tradeoff: replay and reset add work. Benchmark impact remains unverified.

Transaction pooling comparison

Behavior	PgBouncer	PgDog	PGProxyScale preview
Named protocol prepares	Configurable tracking	Rewrites and caches	Ledger replay
SQL PREPARE / EXECUTE	Not in transaction mode	Configurable full mode	Ledger replay
Session SET / RESET	Limited tracked parameters	Tracks session settings	Tracks and replays
LISTEN / NOTIFY	LISTEN needs session mode	Configurable pub/sub	Dedicated backend
Session advisory locks	Need session mode	Pins with parser support	Conservative retention

Sources: official feature and configuration documentation. Details in notes.

Product choice depends on the workload

Product	Technical fit
PgBouncer	Lightweight pooling with well-defined session compatibility limits
PgDog	Pooling with replica routing, sharding and broader session handling
Pgpool-II	PostgreSQL middleware with load balancing and high-availability features
PGProxyScale	Design-partner evaluation of a ledger-based gateway and planned governance

PGProxyScale's proposed differentiation is policy + fairness + state control.
Policy and fairness remain planned capabilities.

Prototype evidence

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Live PostgreSQL 18 compatibility scenarios passed

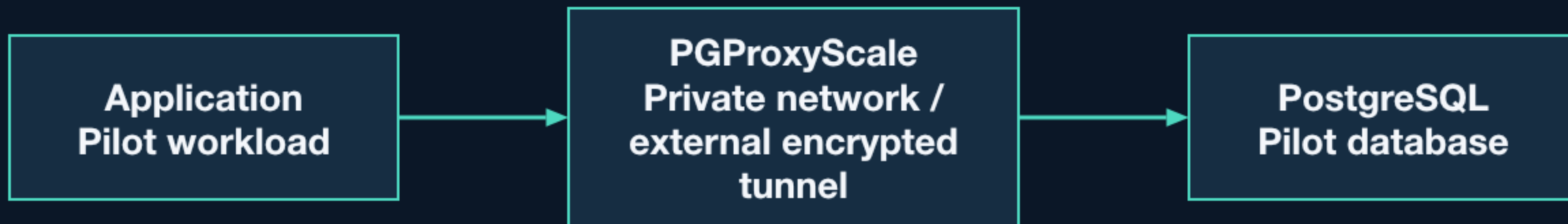
psycopg3 against the native proxy

Prepared statements, session settings, COPY and concurrency

Stateful scenarios can retain a dedicated backend

Evidence scope: functional compatibility. Throughput and fault resilience remain unproven.

Deployment and security boundary



Current: transaction client auth with MD5 or SCRAM, configurable limits

Before production: native TLS, security review, broader driver coverage and fault-injection acceptance.

Development roadmap

Stage	Outcome
Working foundation	Protocol relay, parser, state ledger, authentication and cancellation
Compatibility and production readiness	DDL-safe plans, deeper virtualization, TLS and acceptance coverage
Governance	SQL policy, credentials, principal fairness and cost attribution
Broader gateway	Agent / MCP access, resilience and caching
Optional expansion	Customer-driven sharding

Design-partner pilot

A workload-led evaluation

Select one service with prepared statements and session settings.
Replay its compatibility suite through the proxy.
Measure backend reuse, retention rate and p95 / p99 latency.
Compare against direct PostgreSQL, PgBouncer and PgDog.
Set production gates before expanding traffic.

Pilot deliverable: a compatibility report and a measured go / no-go decision

Sources and evidence

PgBouncer feature map and configuration

www.pgbouncer.org/features.html

www.pgbouncer.org/config

PgDog features, session state, prepared statements and pub/sub

docs.pgdog.dev/features/

docs.pgdog.dev/configuration/pgdog.toml/general/

Pgpool-II product documentation

www.pgpool.net/docs/latest/en/html/intro-what-is.html

PGProxyScale: repository roadmap, ledger code and PostgreSQL 18 live tests

Implementation snapshot: September 29, 2026